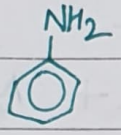
AMINES

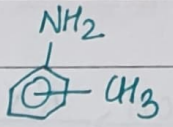
1. Nomenclature ⇒ CH_3NH_2 – methanamine

$CH_3-NH-CH_3$ – N-methylmethanamine

$(CH_3)_3N$ – N,N-dimethylmethanamine.



Aniline



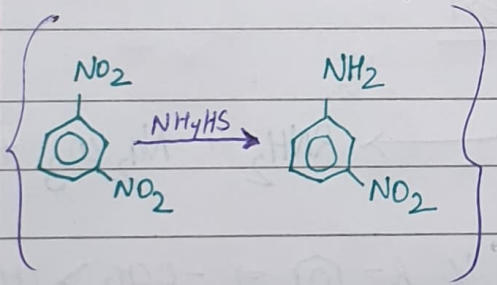
Toluidine

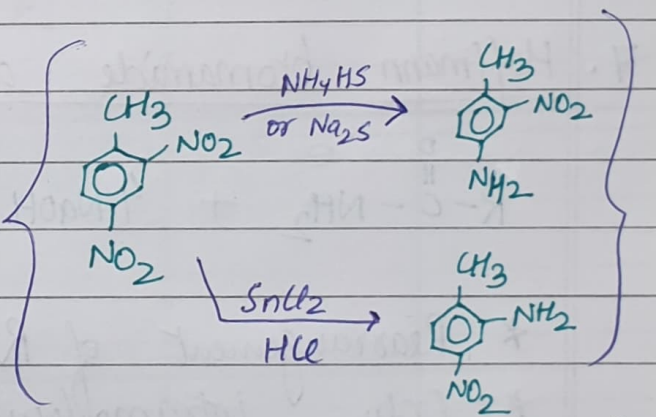
2. NH_3 ← sp^3 → $N(CH_3)_3$ sp^2 $Ni(SiH_3)_3$

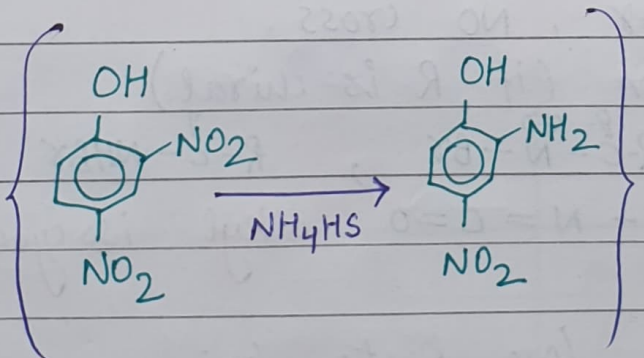
107° 108° 120°

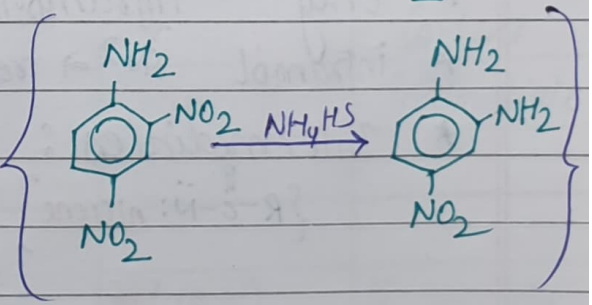
★ ($NCl_3 > NH_3 > NF_3$) B.∠

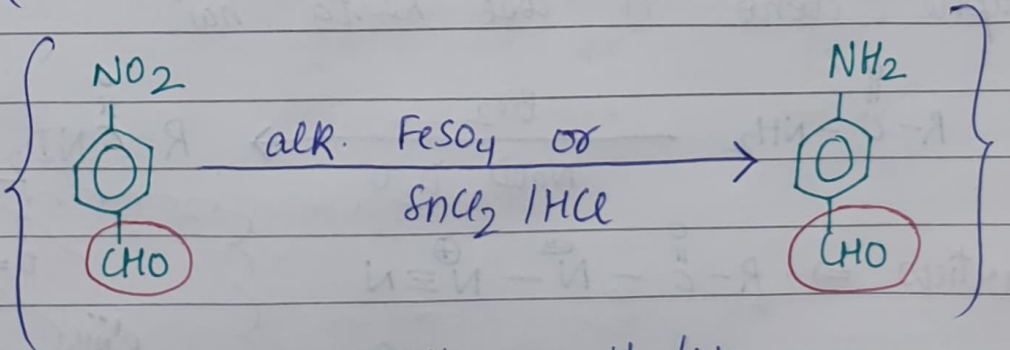
3. RNO_2 —(★ prefer: 1) Fe/HCl, Sn/HCl, Zn/HCl, $SnCl_2$/HCl; 2) H_2/Catalyst; 3) $LiAlH_4$)→ RNH_2









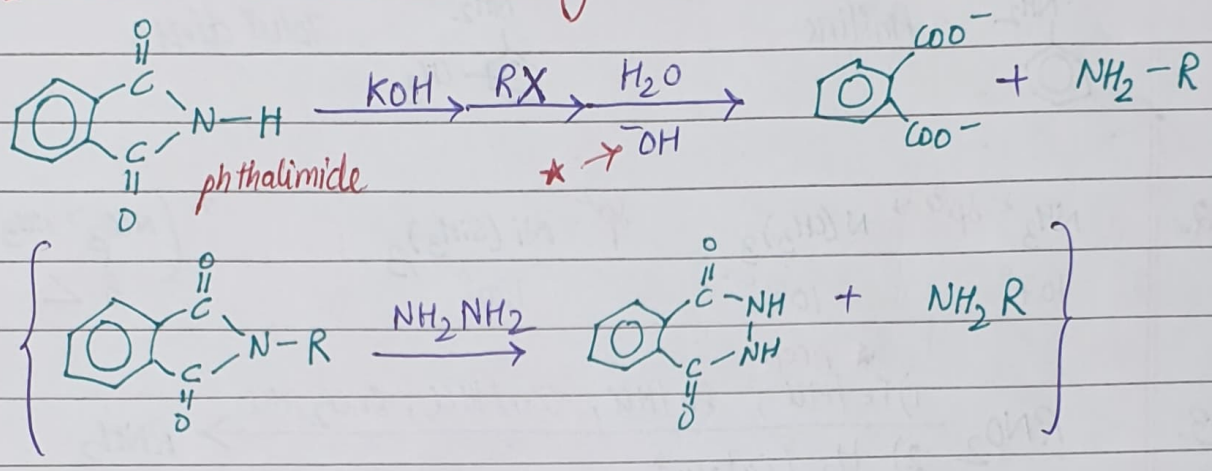


4. $R-C\equiv N$ —($LiAlH_4$ or H_2/Ni or $Na(Hg)/C_2H_5OH$)→ RCH_2NH_2

5. $R-X \xrightarrow{NH_3} RNH_2 \xrightarrow[RX]{RX} R_2NH \xrightarrow{RX} R_3N \xrightarrow{RX} R_4N^{\oplus}$

for $1^{\circ} \Rightarrow NH_3$ (excess)

6. Gabriel's Phthalimide Synthesis.



7. Hoffmann bromamide degradation

$$R-\overset{O}{\overset{\|}{C}}-NH_2 + 4NaOH + Br_2 \longrightarrow RNH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

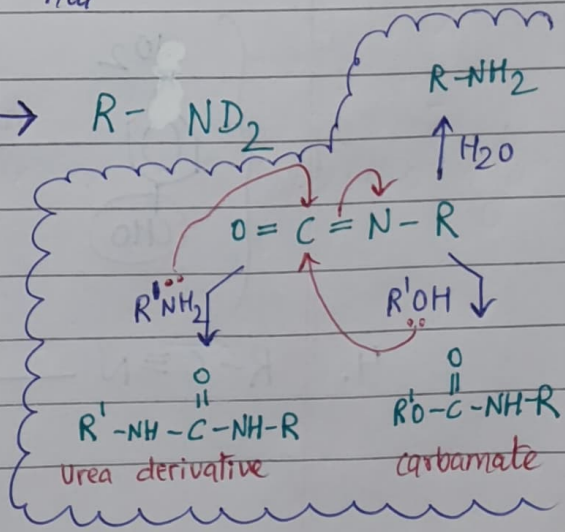
* Rearrangement of R { if $R = C_6H_5 \Rightarrow -OCH_3 > CH_3 > H > Cl > NO_2$ }
* Only intramolecular, NO cross.
* intramol $S_N2 \Rightarrow$ retention (if R is chiral)
* Intermediates: $R-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{N}-Br$, $R-\overset{O}{\overset{\|}{C}}-NHBr$, $R-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{N}H$
  {$R-\overset{O}{\overset{\|}{C}}-\ddot{N}:$ nitrene} $R-N=C=O$ alkyl isocyanate

* N apne dono H loose karta hai

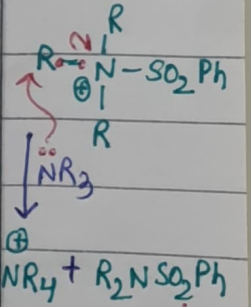
$$R-\overset{O}{\overset{\|}{C}}-NH_2 \xrightarrow[NaOD,\ D_2O]{Br_2} R-ND_2$$

IIdy Curtius $\Rightarrow R-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{N}-\overset{\oplus}{N}\equiv N$

Lossen $\Rightarrow R-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{N}-O-\overset{O}{\overset{\|}{C}}-R$

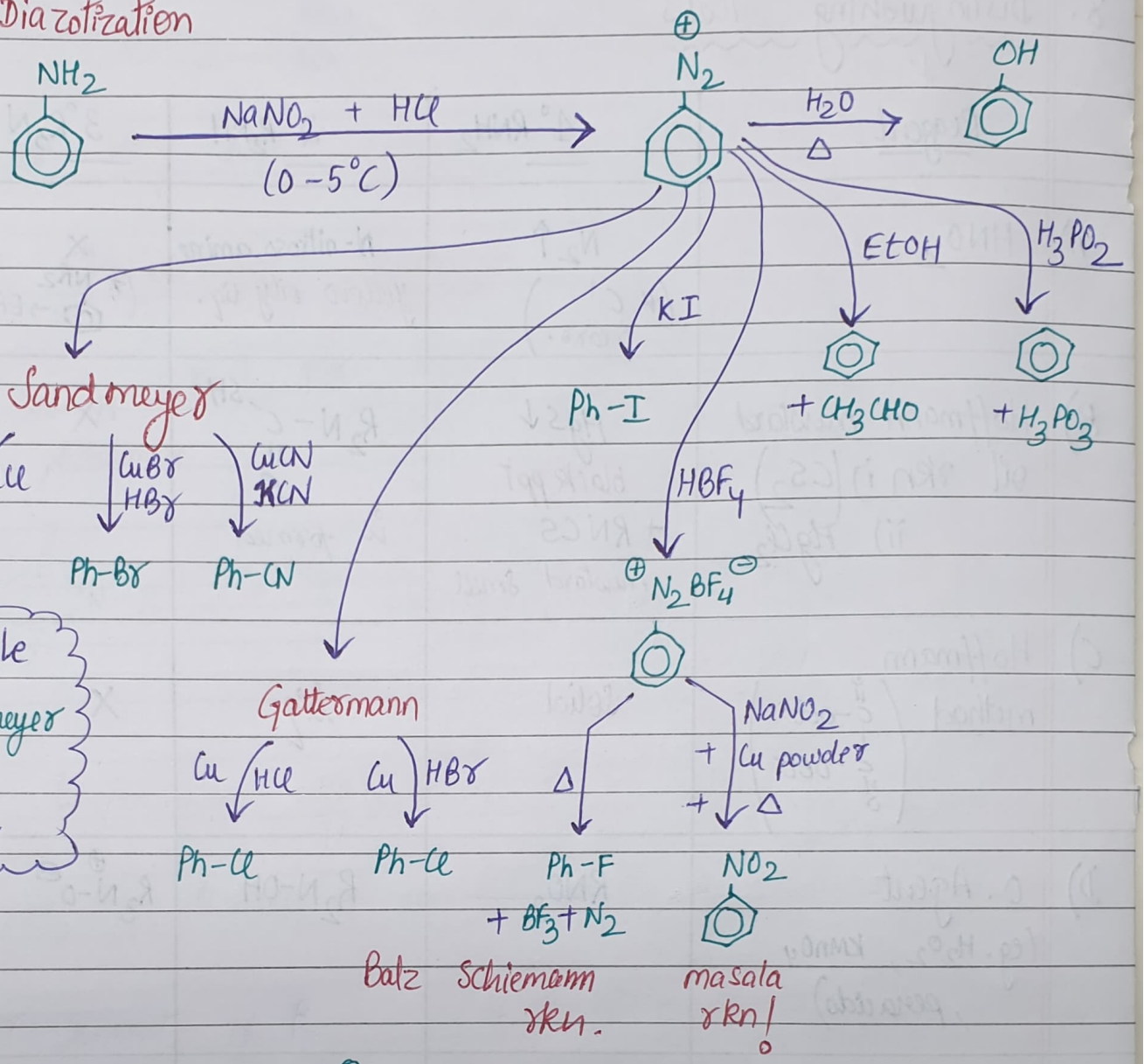


8. Distinguishing Tests

Reagent	1° RNH_2	2° R_2NH	3° R_3N	Remark
A) HNO_2	$N_2\uparrow$ (★ C^+ rearr.)	N-nitroso amine yellow oily liq.	X (★ NR_2 → EAS)	$\overset{\oplus}{NO}$ is $E^{\oplus}$
B) Hoffmann mustard oil rkn i) (CS_2) ii) $HgCl_2$	$HgS\downarrow$ black ppt + RNCS mustard smell	$R_2N-\underset{\underset{S}{\|}}{C}-SH$ is formed	X	
C) Hoffmann method $\left(\begin{matrix} O \\ \| \\ C-OEt \\ \| \\ C-OEt \\ \| \\ O \end{matrix}\right)$	Solid	liq.	X	
D) O. Agent (eg. H_2O_2, $KMnO_4$, peracids)	RNO_2	R_2N-OH	$R_3\overset{\oplus}{N}-\overset{\ominus}{O}$	
E) Carbylamine ($CHCl_3+KOH$)	RNC unpleasent odour	X	X	Carbene intermediate
F) Hinsburg Test ($Ph-SO_2Cl$) Benzene Sulphonyl Chloride	$Ph-SO_2-\overset{\ominus}{N}R$ clear sol^n $\downarrow H^+$ ppt $Ph-SO_2-NHR$	$Ph-SO_2NR_2$ ppt $\downarrow H^+$ xxx	ppt (R_3N) $\downarrow H^+$ clear sol^n ($R_3N\overset{\oplus}{-}H$)	if we take NR_3 excess  $\overset{\oplus}{NR_4} + R_2NSO_2Ph$ $H^+\downarrow$ ppt xxx

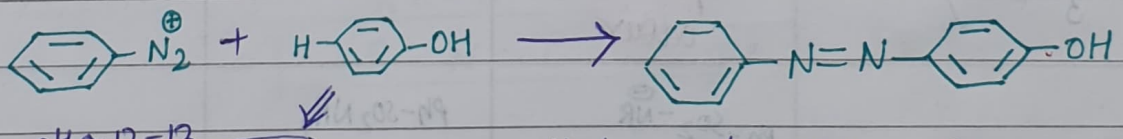
Basics ⇒ 1° ⇒ ✓ $\xrightarrow{alkali}$ Soluble
2° ⇒ ✓ $\xrightarrow{alkali}$ insoluble
3° ⇒ X

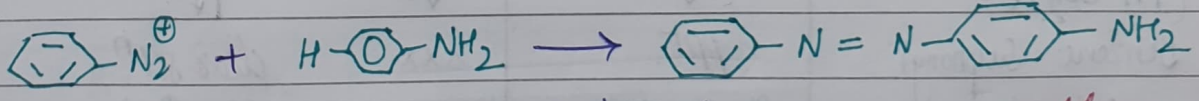
9. Diazotization

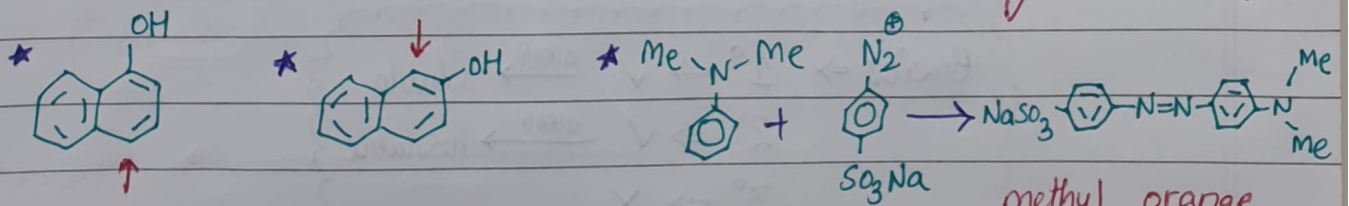


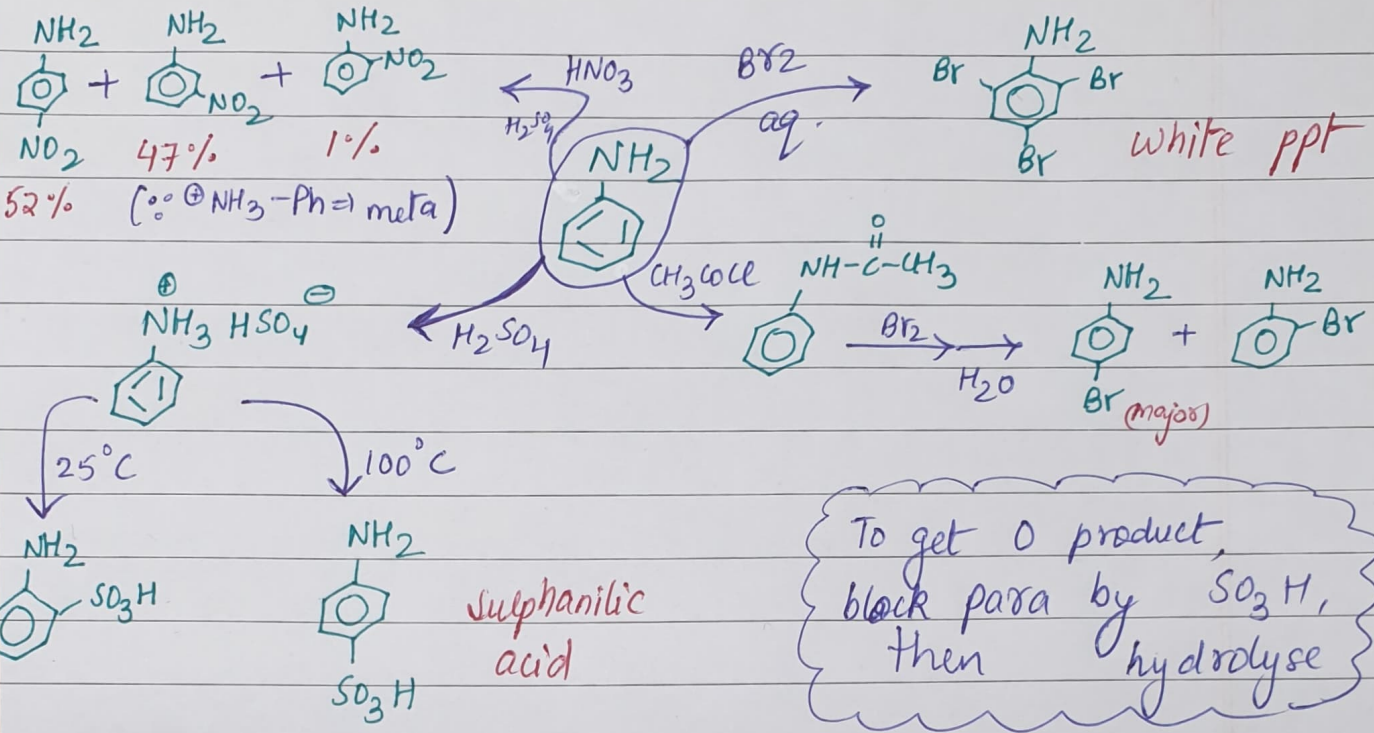
10. Coupling

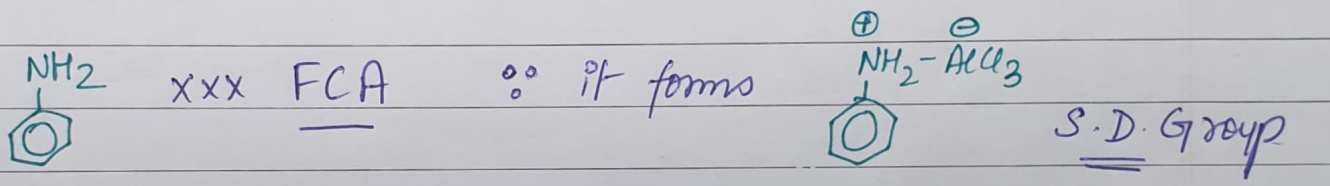
$Ph-\overset{\oplus}{N_2}$ acts as E^+ with  G = NH_2/NRH/NR_2 OH

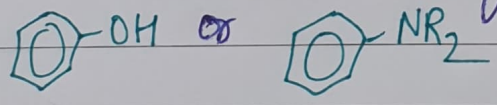






11. 

12. 

13. $\overset{\oplus}{N}O$ is a weak E^+, can only show EAS with ⇒ 

14. $CH_3NH_2 + Ph-\underset{\underset{O}{\|}}{C}-Cl \longrightarrow Ph-\underset{\underset{O}{\|}}{C}-NH-CH_3$

Benzoylation

15. $R-NH_2 + R'-\overset{\overset{O}{\|}}{C}-Cl \longrightarrow R'-\overset{\overset{O}{\|}}{C}-NH-R + HCl$

pyridine is used, which consumes HCl, ∴ eq. forward. →